Day 4 Assignment — Selecting Data

tips.csv + Titanic-Dataset.csv

10 questions. Five on `tips.csv`, five on `Titanic-Dataset.csv`. Submit the notebook with your answers filled in.

Everything here uses only what we did in class: selecting columns, `iloc`, `loc`, boolean indexing, `loc` with a condition, `value_counts()`, and changing values with `loc`.

Nothing from the Bonus section is needed. No isin, no between, no sort_values, no reset_index, no set_index. If you reach for those, there is a simpler way you already know.

Name: ________________________________ Roll no: ________________________

Part A - tips.csv (the restaurant)

A1.

The owner wants to know how generous each customer was, not just how much they left. Make a new column called `tip_percent` - the tip as a **percentage of the bill**, rounded to 1 decimal place. Then show only `total_bill`, `tip` and `tip_percent` for the first 5 rows.

A2.

Answer both parts using **positions only** - no column names anywhere.
1. Print the `total_bill` of the 10th row.
2. Show the last 3 rows, with only the first two columns.

A3.

Now do it with **labels only**. Show rows 20 to 25, with all the columns from `sex` through `size`. Then answer in a comment: how many rows came back, and why is it not 5?

A4.

Two counts for the owner:
1. How many bills were **not** at Dinner? Use `~`.
2. How many bills had a party of 3 or more *and* a tip above 4?

A5.

The owner wants to flag the generous customers. Make a column called `service`. Everyone starts as `'normal'`. Any bill where `tip_percent` is more than 20 becomes `'generous'`. Then count how many of each. (Uses the column you made in A1.)

Part B - Titanic-Dataset.csv

Remember: the column names here start with capital letters - Survived, Pclass, Sex, Age, SibSp, Parch, Fare, Cabin, Embarked.

B1.

How many passengers boarded from each port? Use the `Embarked` column. Then add the three numbers up in a comment. Do they come to 891?

B2.

Two counts:
1. How many **women** were travelling in **third class**?
2. How many of those women survived?

B3.

Some people paid an enormous amount for their ticket. Show only `Name`, `Age` and `Fare` for every passenger who paid more than 200. How many are there?

B4.

Out of all the passengers in **first class**, what percentage survived? Now do the same for **third class**. Put the two numbers next to each other in a comment.

B5.

Make a column called `age_group`. Everyone starts as `'adult'`. Anyone under 18 becomes `'child'`. Then count how many of each.

Think before tomorrow (no code needed). 177 passengers have no `Age` recorded at all. Which group did they land in - and is that the right place for them? Write two lines in a comment saying what you would do about it.

Before the next class

- Re-run the class notebook from top to bottom. Fill in any cell you left blank.
- Run `t.info()` and `t.isnull().sum()` on the Titanic table and look at what comes back.
- Read what `NaN` is, and why `np.nan == np.nan` is **False**.
- Tomorrow: **Day 5 - Data Cleaning.** What to do with a hole.